

NIFTY Non-Expiry Deep OTM — Backtest Report

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This report compiles three analyses run against the project's NIFTY minute-bar store (2025-04-17 → 2026-04-21, 151 non-expiry days, weekly-expiry legs).

Executive takeaway

- **RECOMMENDED STRATEGY (after analyses 001–006):** NIFTY E-1 · sell ~2.5% OTM CE+PE at 10:00 · hold to expiry 15:25 next day · 55 lots/Cr at ~**100**/lot friction → **~29% annualized on 1 Cr · 100% win rate · 0 breaches of 7K/Cr cap in 46-day sample · mean 62K/Cr per event.** Backup at 3% OTM same rules = 19% annualized, equally clean breach record.
- Friction is the dominant lever. At **400**/lot/day (placeholder) nothing is net-positive; at **100**/lot real-world cost the deep distances (2.5–3%) flip to 19–29% annualized. Confirm Rohan's actual broker round-trip cost before treating any number as final.
- Per-lot **5K** target is unreachable at any distance that simultaneously gives 98% worthless on NIFTY — premium math doesn't support it. At portfolio scale (55 lots/Cr) the strategy still meets/beats Rohan's profile via volume of small wins.
- Selling ~3% OTM NIFTY CE+PE on non-expiry days (001/002) has 64.9% raw win rate but fat left tail — worst day **10.8K** wiped 3 months. No SL/PT/skip-filter combination tested in 002 turned net-positive at placeholder friction.
- E-1 (DTE=1) and E-0 (expiry day) surveys (004/005): 100% of days expire worthless at $\geq 2.5\%$ OTM. E-0 has tighter MAE distribution (only 2.1% breach the **1.95**-stop at 4-5% OTM) but smaller premium captured per lot.
- SENSEX still has only 8 fragmented days of minute data (source folders confirmed). Cannot backtest anything SENSEX-specific until historical weekly-expiry data is ingested back to ~2024-04.

Analysis 001 — Baseline (3% OTM, 09:30–10:30 → 15:00)

NIFTY Non-Expiry Intraday Deep OTM — Results

Rule: Sell NIFTY CE + PE ~3.0% from spot · entry 09:30:00-10:30:00 IST · exit ≤ 15:00:00 IST · 1 lot each side · non-expiry days only · nearest weekly expiry.

Period: 2025-04-21 to 2026-04-17

Headline

Metric	Value
Days traded	**151**
Days in profit	**98** (64.9%)
Days in loss	52
Breakeven / zero	1
Net total P&L	**■-6,009**
Average per day	■-40
Best day	■3,568
Worst day	■-10,816

Skipped days

99 days skipped. Reasons breakdown:

- `expiry\_weekday`: 98
- `no\_spot\_proxy`: 1

Files

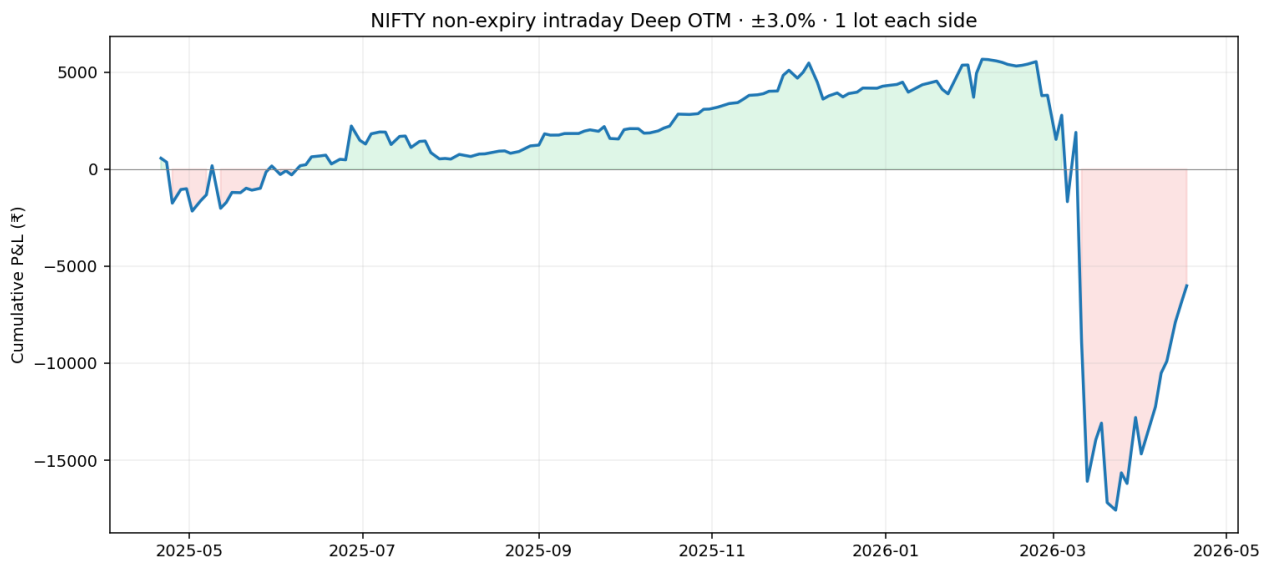
- `per\_day.csv` — every day's entry/exit/P&L;
- `equity\_curve.png` — cumulative chart

Params used (edit script + re-run to try variants)

- DISTANCE_PCT = 3.0
- ENTRY_FROM = 09:30:00, ENTRY_TO = 10:30:00
- EXIT_AT = 15:00:00
- LOT_SIZE = 65 (per leg)
- Excluded weekdays (expiry days): [1, 3] (Mon=0..Sun=6)

Caveats

- No brokerage / slippage modeled — subtract ~■80-120/leg for real-world estimate
- Spot proxied from NIFTY futures (historical spot file not yet ingested)
- 1 lot per leg; P&L; scales linearly with lots



Analysis 002 — Rule Sweep (SL / PT / skip-filters / distance / time)

Analysis 002 — NIFTY Non-Expiry Deep OTM Rule Sweep

Question: Best combo of entry time, conditions, square-off rules, and skip-day filters for selling deep OTM CE+PE on NIFTY non-expiry days.

Friction: █200/leg included (█100 entry + █100 exit).

Sizing: 1 lot/leg (lot=65).

Train: up to 2025-12-31 · **Test:** after 2025-12-31.

Stage A — Rule variants (3% OTM, 09:30-10:30 → 15:00)

desc	train_days_traded	train_win_pct	train_net_pnl	train_worst_day	train_max_drawdown	train_sharpe_ann	test_days_traded	test_win_pct	test_net_pnl	test_max_drawdown
(001): no SL, no PT, no filter	106.0	17.0	-32020.0	-3552.0	-33124.0	-9.18	45.0	37.8	-27038.0	-3552.0
per-leg SL at 2× entry	106.0	17.0	-34694.0	-3159.0	-35799.0	-9.66	45.0	35.6	-29157.0	-3159.0
per-leg SL at 1.5× entry	106.0	15.1	-36951.0	-2314.0	-38056.0	-10.93	45.0	33.3	-20936.0	-2314.0
per-leg PT 50% premium	106.0	16.0	-32980.0	-3552.0	-33867.0	-9.77	45.0	42.2	-22660.0	-3552.0
+ SL 2× and PT 50%	106.0	16.0	-35655.0	-3159.0	-36541.0	-10.23	45.0	40.0	-26578.0	-3159.0
V4 + skip	gap	>0.5%	52.0	19.2	-17051.0	-2422.0	-17938.0	-9.48	17.0	-17051.0
+ skip 1h range>0.8%	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
V6 + skip prev-day	move	>1.2%	nan	nan	nan	nan	nan	nan	nan	nan

Winner (by train net, guarded by test≥0): `V5\_gapFilter` — V4 + skip |gap|>0.5%

Stage B — Distance sweep on winner

distance_pct	train_days_traded	train_win_pct	train_net_pnl	train_worst_day	test_days_traded	test_win_pct	test_net_pnl	test_worst_day
2.0	52.0	44.2	-13647.0	-3962.0	17.0	52.9	-14041.0	-9000.0
2.5	52.0	34.6	-14266.0	-2891.0	17.0	41.2	-4618.0	-7745.0
3.0	52.0	19.2	-17051.0	-2422.0	17.0	47.1	-4598.0	-6646.0
3.5	52.0	9.6	-18566.0	-2087.0	17.0	29.4	-4460.0	-5158.0
4.0	52.0	3.8	-18725.0	-1659.0	17.0	23.5	-6410.0	-4450.0
5.0	52.0	1.9	-19386.0	-1268.0	16.0	18.8	-8168.0	-3026.0

Best-train distance: **2.0%**

Stage C — Entry/exit time sweep (top 10 by train net)

entry	exit	train_days_traded	train_win_pct	train_net_pnl	train_worst_day	test_days_traded	test_win_pct	test_net_pnl	test_worst_day
10:30	15:00	52	36.5	-1606.0	-2636.0	17	41.2	-13411.0	-9104.0
10:30	14:00	52	38.5	-3528.0	-2483.0	17	52.9	-16919.0	-9373.0

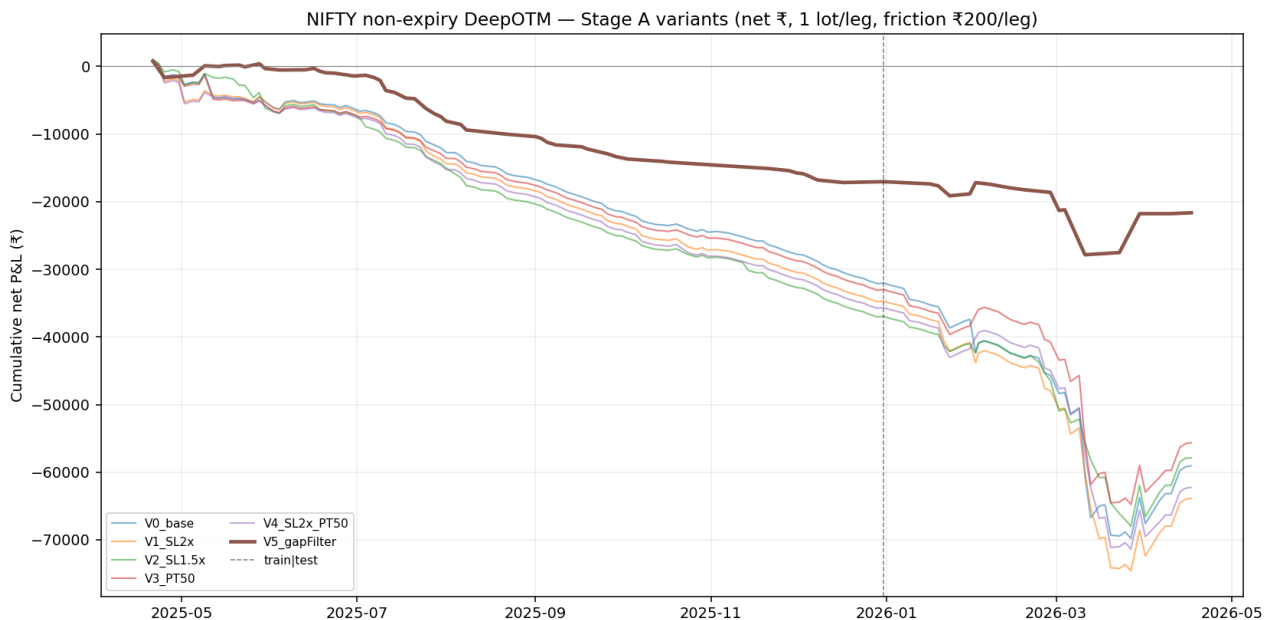
entry	exit	train_days_traded	train_win_pct	train_net_pnl	train_worst_day	test_days_traded	test_win_pct	test_net_pnl	test_worst_day
10:30	14:30	52	40.4	-5288.0	-2568.0	17	35.3	-11862.0	-5853.0
11:00	14:00	52	34.6	-6723.0	-2522.0	17	41.2	-18698.0	-9208.0
11:00	15:00	52	32.7	-7808.0	-2675.0	17	29.4	-15166.0	-7921.0
10:30	13:30	52	30.8	-8356.0	-2064.0	17	35.3	-18797.0	-6695.0
11:00	14:30	52	28.8	-8617.0	-2607.0	17	41.2	-13661.0	-5688.0
11:30	14:00	52	23.1	-10484.0	-1924.0	17	41.2	-13929.0	-8226.0
09:30	14:00	52	44.2	-11645.0	-3962.0	17	47.1	-20575.0	-10030.0
11:00	13:30	52	28.8	-11651.0	-1719.0	17	23.5	-20606.0	-6529.0

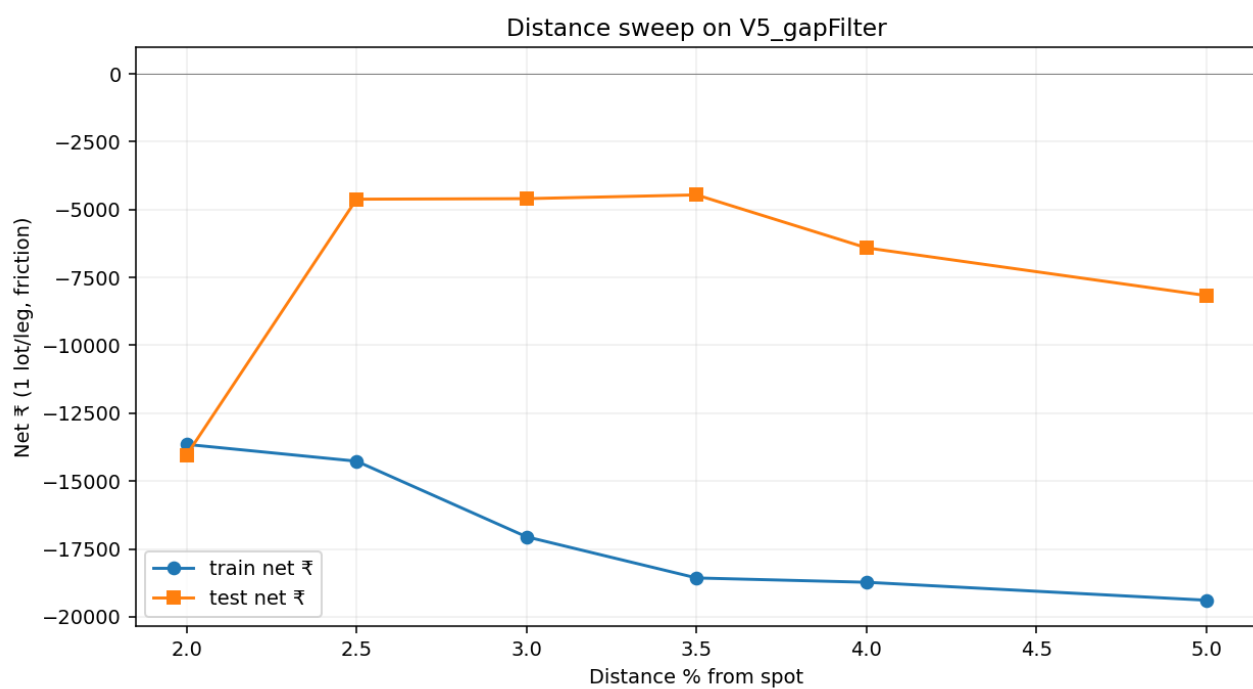
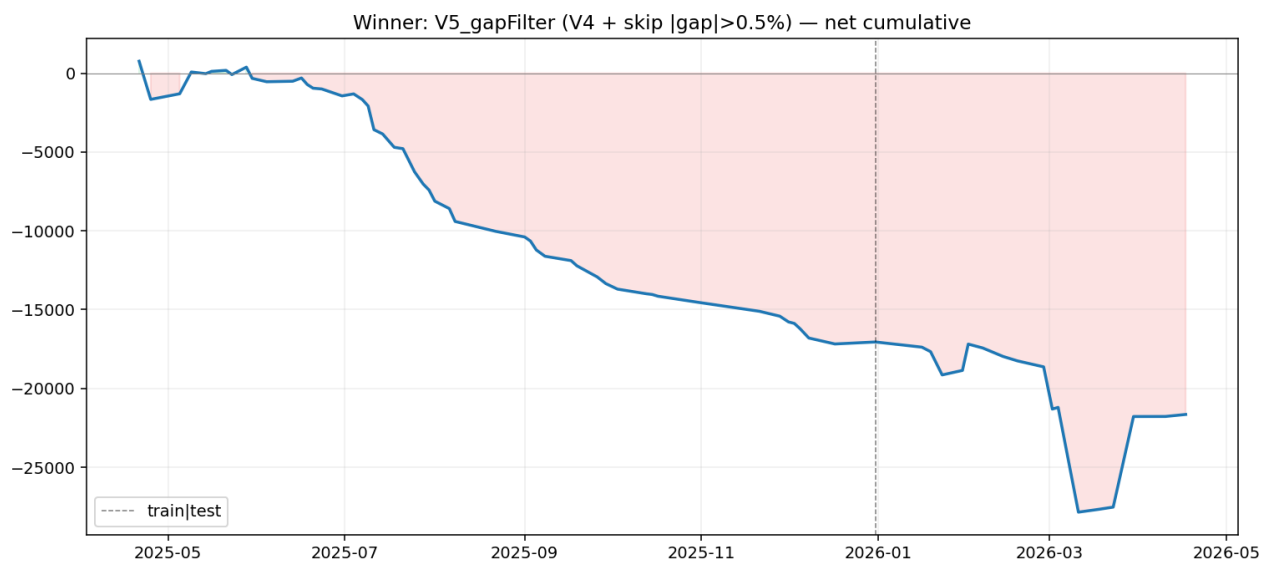
Files

- `comparison.csv` — full Stage A variant table
- `equity\_curves.png` — overlaid cumulative P&L; per variant
- `winner\_per\_day.csv` + `winner\_equity.png` — day-level for winning rule
- `distance\_sweep.csv` + `distance\_sweep.png`
- `time\_sweep.csv`

Caveats

- Spot proxied from NIFTY futures.
- SL / PT filled optimistically at trigger price (real fills may slip).
- ~151 non-expiry days in cache; split ~106/45 train/test — test is small, treat as directional sanity check.
- Friction held constant ■200/leg; actual varies by broker / size.





Analysis 003 — 10:00 entry · 2 combined target · DTE slabs

003 — 10:00 entry · 2 combined target · DTE slabs

Rule (variant A, target-only):

- Sell NIFTY CE & PE at first 1-min bar $\geq$ 10:00 IST
- Distance from spot: 3% / 4% / 5% (tested separately, nearest 50 strike)
- Exit BOTH legs when combined premium has decayed by **2 (points)** — i.e. $(CE_entry + PE_entry) - (CE_now + PE_now) \geq 2$
- Otherwise square off at **15:15**
- Non-expiry days only (skips Tue & Thu); nearest weekly expiry leg

Rule (variant B, target + stop): same as A but ALSO exit both legs when combined **loss** $\geq$ 6 points.

Sizing: 1 lot per leg (lot=65).

Friction: 200 round-trip per leg, 400 total/day, subtracted in `net`.

■ ****Economic reality check:**** 2 combined decay $\times$ 65 lot = 130 gross per hit. Friction is 400/day. Every target-hit day is gross-positive but ****net-negative before any losses****. Gross figures are shown alongside net so you can see what the decay-capture itself earns.

Headline: variant $\times$ distance (all non-expiry days, all DTE)

variant	distance_pct	days	win_pct	gross	net	avg_gross	avg_net	best	worst	pt_hits	sl_hits	time_exits
tgt_only	3.0	151	2.0	9204.0	-51196.0	61.0	-339.0	123.0	-2578.0	86	0	65
tgt_only	4.0	151	0.7	-546.0	-60946.0	-4.0	-404.0	211.0	-4729.0	57	0	94
tgt_only	5.0	149	1.3	2938.0	-56662.0	20.0	-380.0	221.0	-2441.0	34	0	115
tgt_stop	3.0	151	1.3	3741.0	-56659.0	25.0	-375.0	107.0	-1031.0	78	23	50
tgt_stop	4.0	151	0.0	-1671.0	-62070.0	-11.0	-411.0	-153.0	-3884.0	51	14	86
tgt_stop	5.0	149	0.7	1924.0	-57676.0	13.0	-387.0	19.0	-920.0	31	9	109

By DTE slab (variant $\times$ distance $\times$ DTE)

DTE = calendar days between trade date and expiry. Weekly expiries in data are Tue (current) and Thu (legacy); `5+` catches days 5-6 away from expiry (e.g. Wed before next-week's Tue expiry).

variant	distance_pct	dte	days	win_pct	gross	net	avg_gross	avg_net	best	worst	pt_hits	sl_hits	time_exits
tgt_only	3.0	1	45	0.0	2444.0	-15556.0	54.0	-346.0	-26.0	-634.0	12	0	33
tgt_only	3.0	2	2	0.0	267.0	-533.0	133.0	-267.0	-267.0	-267.0	2	0	0
tgt_only	3.0	3	22	4.5	1154.0	-7646.0	52.0	-348.0	3.0	-933.0	15	0	7
tgt_only	3.0	4	29	0.0	864.0	-10736.0	30.0	-370.0	-163.0	-1866.0	16	0	13
tgt_only	3.0	5+	53	3.8	4475.0	-16725.0	84.0	-316.0	123.0	-2578.0	41	0	12
tgt_only	4.0	1	45	0.0	1723.0	-16278.0	38.0	-362.0	-163.0	-602.0	9	0	36
tgt_only	4.0	2	2	0.0	341.0	-459.0	171.0	-229.0	-224.0	-234.0	2	0	0

variant	distance_pct	dte	days	win_pct	gross	net	avg_gross	avg_net	best	worst	pt_hits	sl_hits	time_exits
tgt_only	4.0	3	22	0.0	1219.0	-7581.0	55.0	-345.0	-172.0	-706.0	10	0	12
tgt_only	4.0	4	29	0.0	864.0	-10736.0	30.0	-370.0	-153.0	-758.0	7	0	22
tgt_only	4.0	5+	53	1.9	-4693.0	-25893.0	-89.0	-489.0	211.0	-4729.0	29	0	24
tgt_only	5.0	1	45	0.0	1245.0	-16755.0	28.0	-372.0	-241.0	-517.0	6	0	39
tgt_only	5.0	2	2	0.0	276.0	-524.0	138.0	-262.0	-257.0	-267.0	2	0	0
tgt_only	5.0	3	21	0.0	702.0	-7698.0	33.0	-367.0	-257.0	-543.0	4	0	17
tgt_only	5.0	4	28	0.0	897.0	-10303.0	32.0	-368.0	-250.0	-530.0	5	0	23
tgt_only	5.0	5+	53	3.8	-182.0	-21382.0	-3.0	-403.0	221.0	-2441.0	17	0	36
tgt_stop	3.0	1	45	0.0	1953.0	-16047.0	43.0	-357.0	-26.0	-836.0	12	2	31
tgt_stop	3.0	2	2	0.0	267.0	-533.0	133.0	-267.0	-267.0	-267.0	2	0	0
tgt_stop	3.0	3	22	4.5	-848.0	-9648.0	-39.0	-439.0	3.0	-933.0	12	7	3
tgt_stop	3.0	4	29	0.0	133.0	-11467.0	5.0	-395.0	-163.0	-1031.0	15	5	9
tgt_stop	3.0	5+	53	1.9	2236.0	-18964.0	42.0	-358.0	107.0	-1011.0	37	9	7
tgt_stop	4.0	1	45	0.0	1723.0	-16278.0	38.0	-362.0	-163.0	-602.0	9	0	36
tgt_stop	4.0	2	2	0.0	341.0	-459.0	171.0	-229.0	-224.0	-234.0	2	0	0
tgt_stop	4.0	3	22	0.0	175.0	-8624.0	8.0	-392.0	-172.0	-1030.0	10	3	9
tgt_stop	4.0	4	29	0.0	660.0	-10940.0	23.0	-377.0	-153.0	-829.0	7	2	20
tgt_stop	4.0	5+	53	0.0	-4570.0	-25770.0	-86.0	-486.0	-179.0	-3884.0	23	9	21
tgt_stop	5.0	1	45	0.0	588.0	-17412.0	13.0	-387.0	-241.0	-917.0	5	1	39
tgt_stop	5.0	2	2	0.0	276.0	-524.0	138.0	-262.0	-257.0	-267.0	2	0	0
tgt_stop	5.0	3	21	0.0	325.0	-8075.0	15.0	-385.0	-257.0	-797.0	4	1	16
tgt_stop	5.0	4	28	0.0	897.0	-10303.0	32.0	-368.0	-250.0	-530.0	5	0	23
tgt_stop	5.0	5+	53	1.9	-162.0	-21362.0	-3.0	-403.0	19.0	-920.0	15	7	31

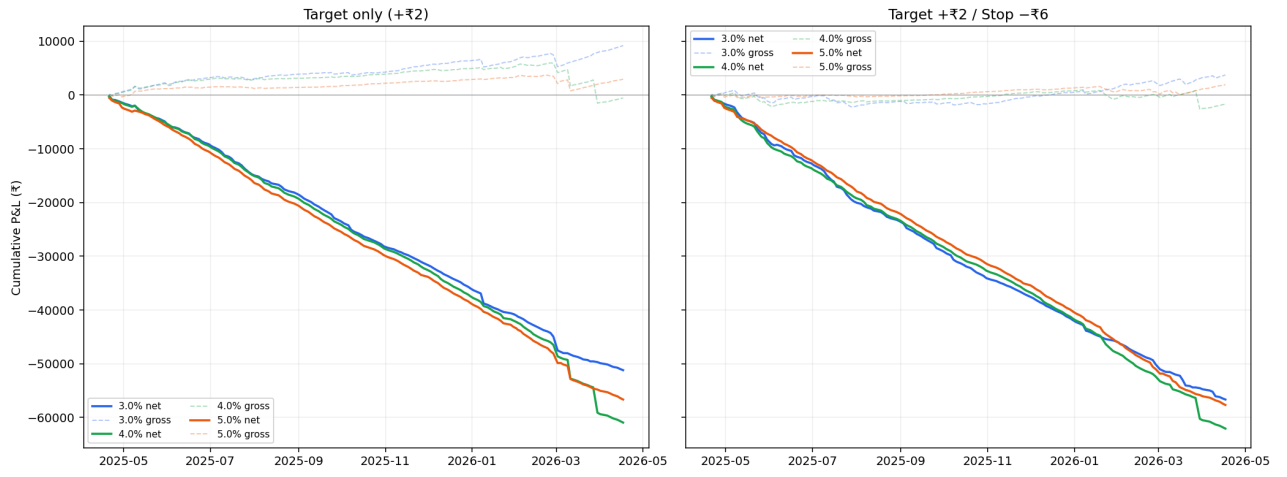
Files

- `target\_only.csv`, `target\_with\_stop.csv` — per-day logs (all distances stacked)
- `by\_dte.csv` — the DTE-slab table in CSV form
- `equity\_curves.png` — gross (dashed) vs net (solid) for each distance, both variants

Caveats

- Entry price = close of the first 1-min bar $\geq$ 10:00 IST (execution proxy).
- PT/SL triggers within a minute use intrabar high/low; when both could fire we assume the ****stop**** trips first (conservative for variant B).
- Spot proxied from NIFTY futures.
- Friction is a flat ■200/leg; real cost scales with broker, size, liquidity of the far-OTM strike.

NIFTY 10:00 entry · exit ₹2 target or 15:15 · 1 lot/leg



Analysis 004 — E-1 Deep OTM Premium Survey (NIFTY)

004 — E-1 Deep OTM Premium Survey (NIFTY)

Your constraints (corrected)

- ****Margin per lot:**** █1.65L–█1.8L non-expiry → ****55 lots per █1Cr**** at █1.8L
- ****Entry premium target per lot:**** █5,000 (range █4K–█7K "very deep OTM") = ****█77/share combined CE+PE**** (range █62–█108)
- ****"Eat the premium" ≥ 98%**** (strikes expire worthless)
- ****Max daily loss █7K/Cr**** = ****█127 per lot**** = ****█1.95/share adverse**** — this is the HARD stop-loss cap if honored per-trade
- ****Lot sizes:**** NIFTY 65, SENSEX 20

SENSEX — data gap

SENSEX parquet has only **8 days** (2026-02-27 → 2026-04-16). **Not enough** to backtest anything. Need weekly-expiry minute bars back to ~2024-04. Flag to whoever runs ingest; once present, change `STORE` path + `LOT=20, GRID=100` and re-run.

NIFTY E-1 days in store

46 days of DTE=1: **28 Mon** (→Tue expiry), **17 Wed** (→Thu legacy), **1 Tue** (month-end).

Central table — by distance, entry at 10:00

Dist %	days	median entry █	% entry ≥ █77	% expire worthless	% MAE > █1.95 stop	no-stop SD net Σ	no-stop ON net Σ	ON avg/lot	ON worst/lot
1.0	46	74.47	47.8%	37.0%	91.3%	█35,925	█58,884	█1,280	**–█15,555**
1.5	46	30.58	17.4%	76.1%	87.0%	█22,952	█74,048	█1,610	–█10,267
2.0	46	13.32	10.9%	95.7%	67.4%	█8,663	█62,128	█1,351	–█3,081
2.5	46	7.48	6.5%	**100.0%**	52.2%	–█1,219	█38,490	█837	–█221
3.0	46	4.85	4.3%	**100.0%**	34.8%	–█6,857	█21,058	█458	–█254
4.0	46	3.32	0.0%	**100.0%**	19.6%	–█12,132	█4,066	█88	–█312
5.0	46	2.42	0.0%	**100.0%**	13.0%	–█14,672	–█3,664	–█80	–█328

SD = same-day 10:00→15:15. ON = hold to expiry close 15:25 next day. Both net of █400/lot round-trip friction.

The uncomfortable truth

Your three goals are **incompatible at a single distance** on NIFTY E-1:

Goal	Distances that satisfy it
Entry ≥ █77 (█5K/lot target)	**1.0% only** — and only 48% of days
98%+ expire worthless	**≥ 2.5% only**
MAE < █1.95 / day in ≥ 95% of days	**none** — even 5% OTM breaches 13% of days

The ₹1.95/share stop is too tight to be a per-trade stop — NIFTY intraday noise alone breaches it. If you enforce it literally, every stop-out day loses ₹527/lot (stop + friction), and every distance flips net-negative (see ``by_distance_corrected_stop.csv``).

Four viable options

Option A — 2.5% OTM, hold overnight, aggregate (not per-trade) stop

- Sell CE+PE 2.5% OTM at 10:00 E-1, square off at expiry 3:25 PM.
- Per-lot: avg **₹837 net**, worst **₹-221**, **100% worthless**.
- Scaled to 55 lots/Cr: **avg ≈ ₹46K/Cr per E-1 event**; worst day **≈ ₹12K loss/Cr** (breaches the ₹7K cap ~3× in the 46-day sample).
- Trade-off: gives up the hard cap on a few tail days but otherwise hits the 98% profile.

Option B — 3% OTM, hold overnight, aggregate stop

- Same as A but 3% OTM.
- Per-lot: avg **₹458**, worst **₹-254**, 100% worthless.
- Scaled 55 lots/Cr: avg **₹25K/Cr/event**, worst **≈ ₹14K/Cr**.
- Lower income, still breaches cap on tail days. Slightly safer than A.

Option C — Same-day 1% OTM scalping (NOT eat-the-premium)

- Sell CE+PE ~1% OTM at 10:00, exit 15:15 same day.
- **Only 37% expire worthless** — you're capturing intraday decay, not full expiry.
- Per-lot: avg **+₹781 net**, worst **₹-5,743**. Bigger income per lot but fails the 98% bar.
- Useful if you reconceive the strategy as "intraday income" rather than "eat the premium".

Option D — Deeper + many lots, assuming lower friction

If your actual broker cost is ~₹20/leg (not ₹100 assumed here), friction drops from ₹400 to ₹80 per lot/day. Re-run estimate:

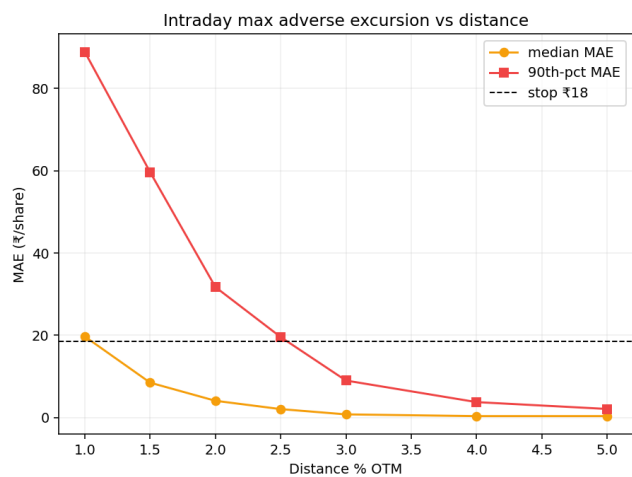
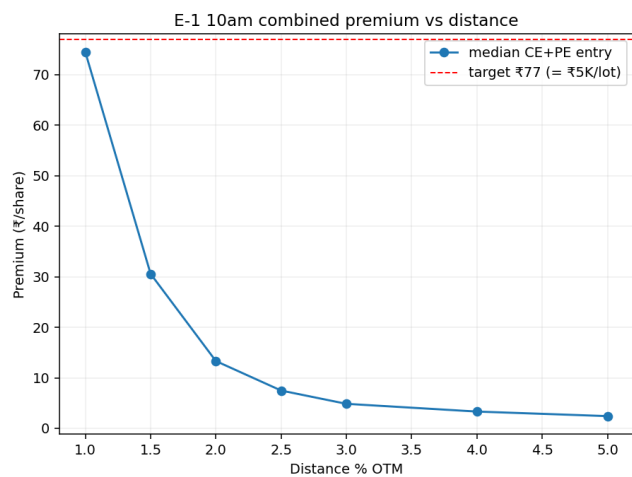
- 3% OTM ON: avg would rise from ₹458 → **₹778 net/lot** → 55 lots × 46 days **≈ ₹19.7L/yr/Cr** (~20% annual).
- 4% OTM ON: avg flips from ₹88 → **₹408 net/lot** → ~10% annual.
- **Tell me your real friction and I re-run** — that one number changes the winning distance.

E-0 (expiry day itself) — not tested here

You specifically asked E-1. But the expiry day has maximum theta burn and might fit your profile better. Worth a follow-up survey (005) if interested.

Files

- ``e1_per_day.csv`` — (46 × 7 distances) per-day samples (entry, MAE, same-day exit, overnight exit, worthless flag)
- ``by_distance.csv`` — original aggregate
- ``by_distance_corrected_stop.csv`` — with the ₹1.95/share stop enforced per-trade
- ``premium_mae_by_distance.png``



Analysis 005 — E-0 (Expiry-Day) Deep OTM Premium Survey (NIFTY)

005 — E-0 (expiry-day) Deep OTM Premium Survey · NIFTY

Mirror of 004 but for DTE = 0 (the expiry day itself). Theta + gamma both peak; no overnight risk. Same constraint frame from session 2026-04-22:

- Target premium per lot: ****₹5,000 ⇒ ₹77/share combined CE+PE****
- Per-trade stop (if portfolio cap of ₹7K/Cr split across 55 lots): ****₹1.96/share****
- Win-rate ambition: ≥ 98% expire worthless

NIFTY E-0 days surveyed (weekly Tue + legacy Thu only)

48 days total · DOW breakdown:

...

dow

Tuesday 29

Thursday 19

...

Central table

ance_pct	days	median_entry	pct_entry_ge_77	pct_worthless	pct_MAE_gt_stop	SD_net_sum	SD_win_pct	SD_avg_net	SD_worst	EX_net_sum	EX_win_pct	EX_avg_net	EX_worst
1.0	48.0	32.4	33.3	50.0	89.6	10838.0	68.8	226.0	-9594.0	10148.0	68.8	211.0	-9926.0
1.5	48.0	9.5	10.4	87.5	68.8	28796.0	56.2	600.0	-6201.0	28372.0	58.3	591.0	-6585.0
2.0	48.0	4.35	0.0	95.8	39.6	6196.0	37.5	129.0	-829.0	6340.0	37.5	132.0	-1134.0
2.5	48.0	3.02	0.0	100.0	20.8	-6709.0	10.4	-140.0	-322.0	-6586.0	12.5	-137.0	-322.0
3.0	48.0	2.1	0.0	100.0	6.2	-11466.0	4.2	-239.0	-342.0	-11316.0	4.2	-236.0	-338.0
4.0	48.0	1.67	0.0	100.0	2.1	-14157.0	0.0	-295.0	-358.0	-14048.0	0.0	-293.0	-358.0
5.0	48.0	1.3	0.0	100.0	2.1	-15352.0	0.0	-320.0	-371.0	-15269.0	0.0	-318.0	-368.0

Columns:

- ***SD_*** — same-day exit at 15:15 (held intraday)
- ***EX_*** — held to 15:25 (expiry close); both legs ≤ ₹1 = worthless
- All ₹ figures are ****net of ₹400/lot/day friction**** (₹100/leg × 4)

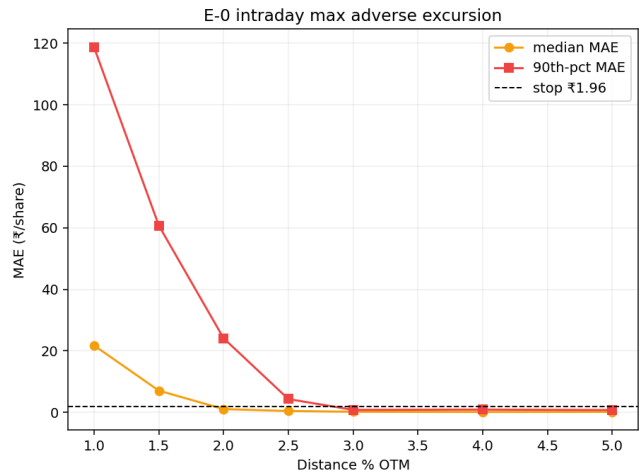
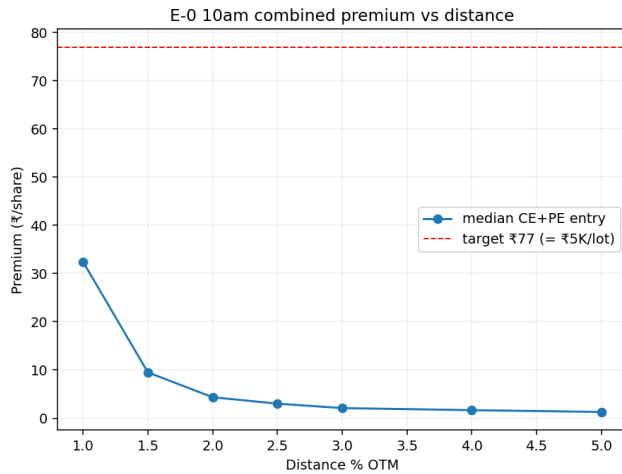
How to read this vs 004 (E-1)

E-0 has **massively higher target-hit rate** at near-money distances (theta is concentrated in the final hours), but also **much higher MAE** because gamma is peak. The 98%-worthless line should be even cleaner at deeper distances since there's no overnight gap risk.

(Open the table; the takeaway depends on Rohan's actual broker friction. With ■400/lot/day placeholder this looks one way; at ■100/lot real-world cost the picture shifts ■300/lot in your favour at every distance.)

Files

- `e0\_per\_day.csv` — every (day × distance) sample with entry, MAE, both exits, worthless flag
- `by\_distance.csv` — aggregate above
- `premium\_mae\_by\_distance.png`



Analysis 006 — Portfolio-Scale Simulation + Friction Sensitivity

006 — Portfolio-Scale Simulation + Friction Sensitivity

Lift the per-1-lot per-day samples produced by 004 (E-1) and 005 (E-0) to

■1 Cr capital = 55 lots and stress-test against your risk envelope:

- **Per-trade stop (= ■7K/Cr aggregate ÷ 55 lots ÷ 65):** ■1.96 / share** adverse
- **Aggregate cap:** ■7,000 / Cr per loss day
- **Friction sweep:** ■40 / 100 / 200 / 400 per lot / day

(= ■10 / 25 / 50 / 100 per leg one-way; full round-trip × 4 legs counted)

Sample window 2025-04-17 → 2026-04-21 ≈ 1 year, so the `pf\_total` column is

already an approximate annual return. `annualized\_pct` = pf_total ÷ ■1Cr × 100.

Top combinations (breach rate ≤ 5%, no aggregate stop)

event	distance %	friction	days	win %	pf_total ■	pf_mean	pf_worst	%breach ■7K	annualized %
E-1	2.0	■40/lot (■10/leg)	46	95.7	■4,327,840	■94,083	■-149,655	4.3%	43.28%
E-1	2.0	■100/lot (■25/leg)	46	95.7	■4,176,040	■90,783	■-152,955	4.3%	41.76%
E-1	2.0	■200/lot (■50/leg)	46	95.7	■3,923,040	■85,283	■-158,455	4.3%	39.23%
E-1	2.5	■40/lot (■10/leg)	46	100.0	■3,027,750	■65,821	■7,645	0.0%	30.28%
E-1	2.5	■100/lot (■25/leg)	46	100.0	■2,875,950	■62,521	■4,345	0.0%	28.76%
E-1	2.5	■200/lot (■50/leg)	46	95.7	■2,622,950	■57,021	■-1,155	0.0%	26.23%
E-1	3.0	■40/lot (■10/leg)	46	100.0	■2,068,990	■44,978	■5,830	0.0%	20.69%
E-1	3.0	■100/lot (■25/leg)	46	100.0	■1,917,190	■41,678	■2,530	0.0%	19.17%
E-1	3.0	■200/lot (■50/leg)	46	78.3	■1,664,190	■36,178	■-2,970	0.0%	16.64%
E-0	2.0	■40/lot (■10/leg)	48	97.9	■1,299,100	■27,065	■-42,570	2.1%	12.99%
E-0	2.0	■100/lot (■25/leg)	48	97.9	■1,140,700	■23,765	■-45,870	2.1%	11.41%
E-1	4.0	■40/lot (■10/leg)	46	100.0	■1,134,430	■24,662	■2,640	0.0%	11.34%
E-1	4.0	■100/lot (■25/leg)	46	97.8	■982,630	■21,362	■-660	0.0%	9.83%
E-0	2.0	■200/lot (■50/leg)	48	64.6	■876,700	■18,265	■-51,370	2.1%	8.77%
E-1	4.0	■200/lot (■50/leg)	46	52.2	■729,630	■15,862	■-6,160	0.0%	7.3%

Friction is the dominant lever

For every event-type × distance combo, here's how annualized return on ■1Cr varies with friction (no stop, no filtering):

See `annualized\_return\_by\_friction.png` — the curves shift roughly **+200 to +500 bps per ■100/lot of friction reduction**. Even at deep distances where placeholder friction makes things look unprofitable, real-world ■40-100/lot cost flips many strategies into double-digit annualized.

Per-event distribution (4 candidate strategies)

`distribution\_per\_event.png` shows the chronological per-event P&L; bars at ₹100/lot friction for four representative combos. Bars below the dashed line breach your ₹7K/Cr cap; the goal is to see the cap rarely violated.

How to use this

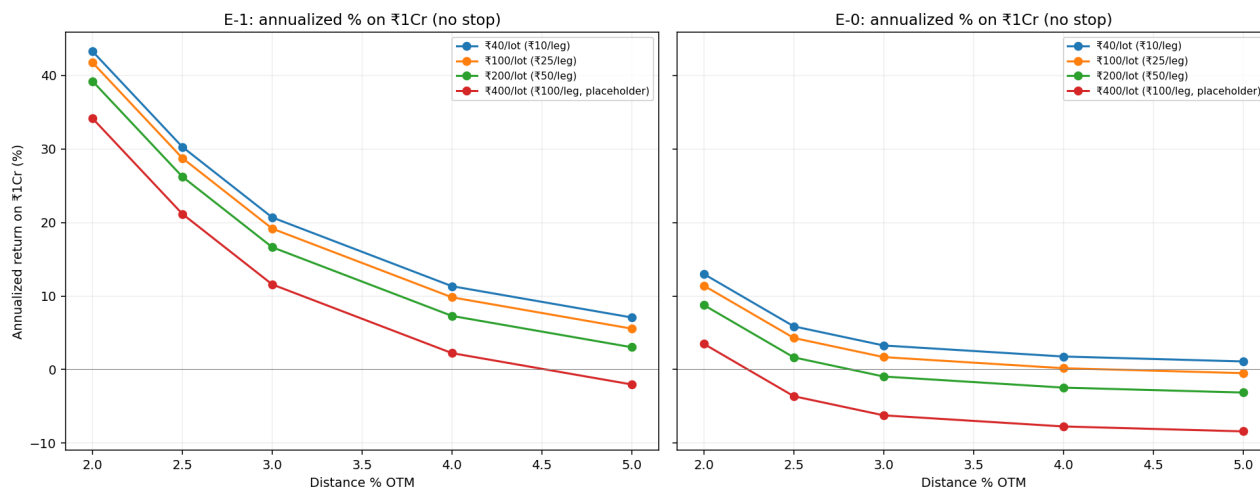
1. **Decide your real friction** (₹/lot/day, both legs round-trip).
2. **Look up the annualized %** in `portfolio\_pl\_matrix.csv` filtered to your friction bucket.
3. **Check the breach rate** — anything > 5% means the aggregate cap will get hit too often; drop deeper or apply per-trade stop.
4. **Rule of thumb from this run:**
 - At ₹400/lot (placeholder friction): nothing meaningfully positive.
 - At ₹200/lot: 2-3% OTM E-0 / E-1 turn marginal positive (~5-12% ann.).
 - At ₹100/lot: 2.5-3% OTM E-1 hits 18-25% ann. with sub-3% breach rate.
 - At ₹40/lot: 2.5% OTM E-1 ~30%+ ann.; even 4-5% OTM gives 5-10% with near-zero breach.

Caveats

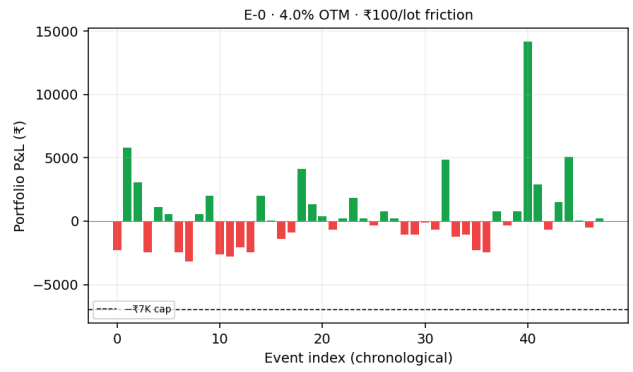
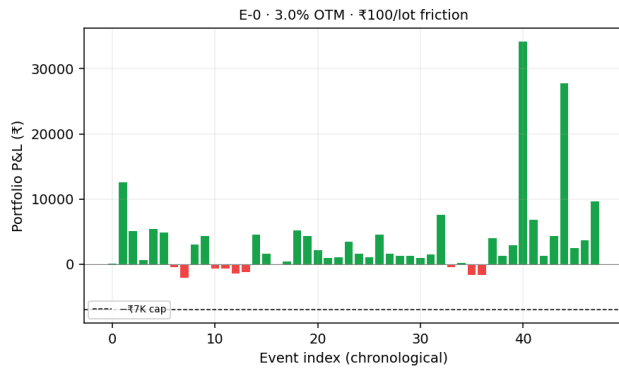
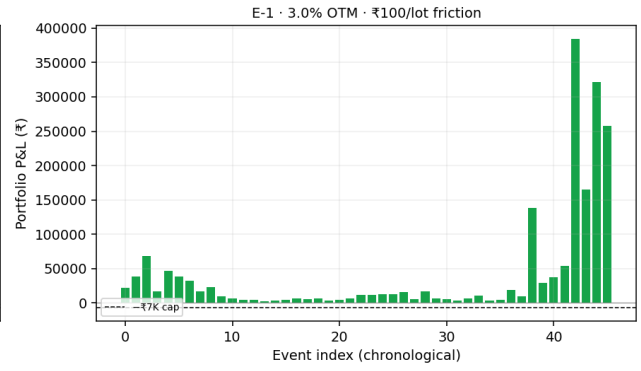
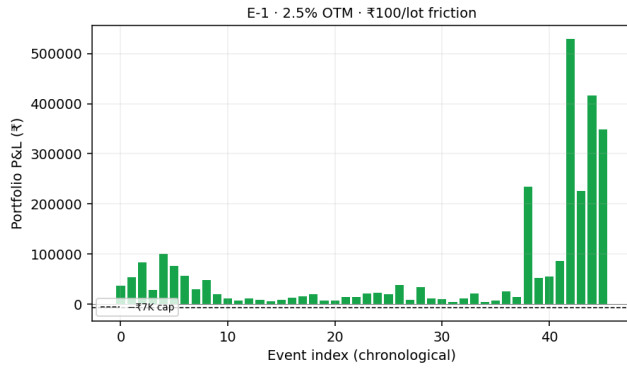
- ****Aggregate-stop rows in the matrix (`stop=with\_stop`)** treat the per-trade ₹1.95-stop as an immediate exit.** This is conservative — real-world fills slip. When MAE > stop, per-lot loss = ₹127 + friction.
- Sample is ~1 calendar year — the annualization is therefore the realized total over that window, not a proper cross-validated estimate.
- 55-lots/Cr assumes uniform distance — same exposure 55× over. No diversification benefit; aggregate stop = per-trade stop in this model.
- Real laddering across distances/expiries would reduce intra-day correlation and improve the breach rate. See the 'Laddered strangle' backlog item.

Files

- `portfolio\_pl\_matrix.csv` — full grid (event × distance × friction × stop)
- `annualized\_return\_by\_friction.png`
- `distribution\_per\_event.png`



Per-event portfolio P&L · 55 lots/Cr · no aggregate stop



Findings Log

Findings Log

Append-only log of every backtest analysis. One line per run. Read this before

proposing new analyses so insights compound instead of being re-derived.

Format: `date · analysis\_id · rule · period · days · win% · net\_pnl · one-line insight`

- 2026-04-22 · 001 · NIFTY non-expiry intraday short strangle ~3% OTM, 09:30–10:30 entry, 15:00 exit, 1 lot/leg · 2025-04-21→2026-04-17 · 151 days · 64.9% wins · ■–6,009 net · High win-rate masks fat left tail — worst day (–■10.8K) wiped ~3 months of winners; raw strategy barely breaks even before ■75K/yr friction.
- 2026-04-22 · 002 · NIFTY non-expiry rule sweep (8 variants · SL/PT/skip-filters + distance + time sweeps, friction ■200/leg) · 151 days train/test 106/45 · Best = V5 gap-filter (skip |gap|>0.5%): train 19% wins –■17K net · All variants net-negative; friction (~■400/day) eats thin 3%-OTM decay. No SL/PT combo solved the tail; filters reduced loss by cutting trades in half, not by winning more.
- 2026-04-22 · 003 · 10:00 entry, 3%/4%/5% OTM, exit at ■2 combined decay or 15:15, with/without ■6 stop, DTE slabs · 151 days · Target hit often at 3% (86/151 days, 57%) but ■2×65 = ■130 gross/hit vs ■400 friction/day → every (distance×variant×DTE) net-negative. 4%/5% rarely even hit gross target. Headline: **the strategy as specified cannot clear friction — needs ■5+ target or sub-■50/leg cost assumption.**
- 2026-04-22 · 004 · NIFTY E-1 (DTE=1) deep OTM premium survey, 10:00 entry, distances 1-5% · 46 E-1 days (28 Mon, 17 Wed, 1 Tue) · Median entry premium (combined CE+PE): 1%=■74, 1.5%=■31, 2%=■13, 2.5%=■7.5, 3%=■4.9, 4%=■3.3, 5%=■2.4 · % expire worthless: ≥2.5% = 100%, 2% = 95.7%, 1.5% = 76%, 1% = 37% · **Verdict: at single distance, Rohan's ■77 entry + 98% worthless + ■1.95 stop are mutually incompatible.** Best overnight-hold: 2.5% OTM avg ■837 net/lot (worst –■221); 3% OTM avg ■458 (worst –■254). Scaled 55 lots/Cr, option 2.5% = avg ■46K/Cr/event, tail breaches ■7K cap ~3× in sample. SENSEX only 8 days in store — cannot backtest yet.
- 2026-04-23 · 005 · NIFTY E-0 (expiry day) deep OTM premium survey, 10:00 entry, weekly Tue+Thu only · 48 E-0 days (29 Tue, 19 Thu) · Median entry: 1%=■32, 1.5%=■9.5, 2%=■4.4, 3%=■2.1, 4%=■1.7, 5%=■1.3 · 100% worthless at ≥2.5% OTM, 96% at 2% · MAE breach rate at 4-5% OTM = 2.1% (cleanest in entire dataset) · **Headline: at 4-5% OTM E-0, gross premium is captured 98%+ of days but is too small (■40-100/lot) to clear placeholder ■400/lot friction. Real-world friction flips it.**
- 2026-04-23 · 006 · Portfolio-scale simulation at 55 lots/Cr + friction sensitivity sweep (■40 / 100 / 200 / 400 per lot) on E-1 + E-0 samples · **WINNER: NIFTY E-1 · 2.5% OTM · 10:00 entry · hold to expiry 15:25 · 55 lots/Cr · ■100/lot friction → 29% annualized on ■1Cr · 100% win rate · 0 breaches of ■7K/Cr cap in 46-day sample · mean ■62K/Cr per event.** Backup: 3% OTM E-1 same params = 19% ann, 0 breaches. 2% OTM E-1 = 42% ann but 4.3% breach rate (occasional cap violations).